Grocery Delivery Management System

System Requirements Documentation

Susan Rogers

SP26-NW-INFO-C450-24493

Jafrina Jabin

03/09/2026

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Grocery Delivery Management System

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Problem Statement

Grocery shopping is necessary but time-consuming for many people, including busy professionals, elderly individuals, families, and customers with limited mobility. Traditional shopping requires travel, time, and physical effort. Many local grocery stores do not have an affordable way to offer online ordering and delivery. This leads to inaccurate stock information, delayed deliveries, and poor communication with customers.

Objectives of the System

Provide an easy-to-use grocery delivery platform for customers and local grocery stores. Reduce manual errors, improve inventory accuracy, streamline order fulfillment, and increase customer satisfaction.

System Functionality Overview

Customers browse and search products, add items to a cart, place delivery orders, and track order status. Managers maintain product listings and inventory, review orders, and assign deliveries. Delivery staff update delivery progress so customers can see real-time status.

System Requirements

• Register and securely log in
• Browse products by category and search by keywords
• Add items to a cart and place delivery orders
• View order history and delivery status
• Receive notifications for confirmations and updates
• Managers add, update, or remove products
• Inventory automatically updates when orders are placed
• Assign delivery orders and update delivery status

Typical Customers

Customers ordering groceries for home delivery, grocery store managers, and delivery personnel.

Project Planning (Software, Hardware, Network)

Software: Java application developed in NetBeans using in-memory data storage

Development Approach

The system was developed in Java using Swing for the interface and Java ArrayLists in the DataStore class for in-memory data management

Project Plan

1. Weeks 1–2: finalize requirements,design system data model
2. Weeks 3–4: implement user registration and login
3. Weeks 5–7: implement product management, inventory, and order placement
4. Week 8: testing and midterm demo
5. Weeks 9–11: implement delivery assignment and tracking
6. Weeks 12–14: testing, bug fixes, and improvements
7. Week 15: final testing and presentation

Estimated Cost and Business Value

The primary costs involve development time and basic hosting services. The system provides strong business value by enabling local grocery stores to operate online, improving efficiency, reducing errors, and increasing customer satisfaction.

Grocery Delivery Management System

System Design Overview and Communication Document

Susan Rogers

1. Customer Problem Statement

Purchasing groceries is an essential activity but sometimes requires a significant investment of time and effort. Customers such as working professionals such as myself, older adults, households with young children, and people who have disabilities most times, experience difficulty traveling to stores, navigating aisles, and waiting in checkout lines. Although online grocery services are available through large retailers, many small and local grocery stores don't have cost-effective systems to support online ordering and delivery.

From a customer's perspective, an effective grocery delivery system should make it possible to look for products, submit orders remotely, and receive dependable updates about delivery progress. Customers expect that the exact product availability is accurate and that deliveries occur within a reasonable timeframe. Store managers, on the other hand, expect a system that makes work, minimizes inventory discrepancies, and improves coordination between ordering and delivery. Overall, the end goal of this system is to improve life overall for customers while helping local grocery stores operate more efficiently.

2. Glossary of Terms

Customer: An individual who selects grocery items and submits delivery orders

Manager: A store representative responsible for overseeing products, inventory, and orders

Delivery Staff: Employees tasked with delivering grocery orders to customers

Product: A grocery item offered for purchase within the system

Inventory: The available quantity of products maintained by the store

Order: A completed request for grocery items to be delivered

Order Status: An indicator representing the current stage of an order

Notification: A system-generated message informing customers of order updates

3. System Requirements

3.1 Functional Requirements

- Users shall be able to create accounts and authenticate securely.

- Customers shall be able to view products organized by category.

- Customers shall be able to locate products using a search function.

- Customers shall be able to select items and store them in a shopping cart.

- Customers shall be able to submit orders for grocery delivery.

- Customers shall be able to review previously placed orders.

- Customers shall be able to monitor the progress of active deliveries.

- Customers shall receive system notifications related to their orders.

- Managers shall be able to create, update, and remove product listings.

- The system shall automatically adjust inventory when orders are placed.

- Managers shall be able to view and manage incoming orders.

- Managers shall be able to assign delivery tasks to delivery staff.

- Delivery staff shall be able to update the status of assigned deliveries.

3.2 Nonfunctional Requirements (FURPS)

- Functionality: The system shall support the accuracy of orders and inventory.

- Usability: The system should be easy to use and get to for users with varying technical skills.

- Reliability: The system shall consistently maintain correct and up-to-date information.

- Performance: System responses shall occur within a small amount of time limits.

- Supportability: The system should be structured to allow flexibility for upgrades.

4. Plan of Work

Project development followed the planned schedule established during the proposal phase. Initial efforts focused on preparing the development environment and refining system requirements. Subsequent planning addressed core features including user access, product management, inventory handling, and order processing. Additional work is planned for delivery coordination, system testing, and final refinements before project completion.

5. When to Start Coding

Implementation starts after completing requirements analysis and system design activities. Development initially concentrated on the server-side application logic, followed by user-facing components. Data storage integration and testing were conducted alongside development to ensure the system runs smoothly and that data remains consistent within the in-memory DataStore.

Use Case Diagram

The following use case diagram illustrates how different actors interact with the Grocery Delivery Management System. It shows primary system functionalities and shows how customers, store managers, and delivery staff interact with the system through required and optional behaviors.

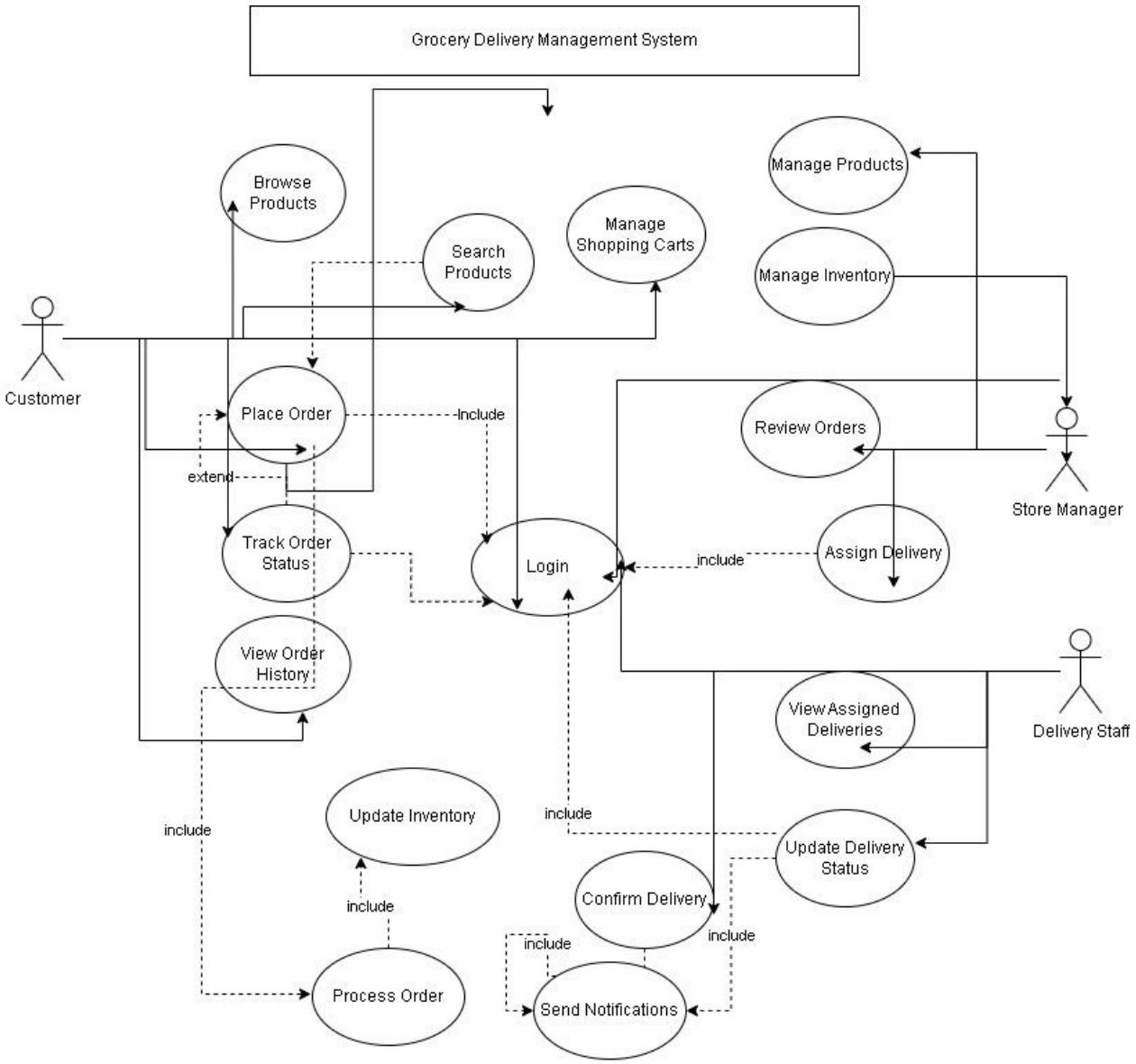


Figure: Grocery Delivery Management System Use Case Diagram

Class Diagram

The following class diagram shows the main classes used in the Grocery Delivery Management System and how they relate to one another. It highlights users, orders, products, and delivery components.

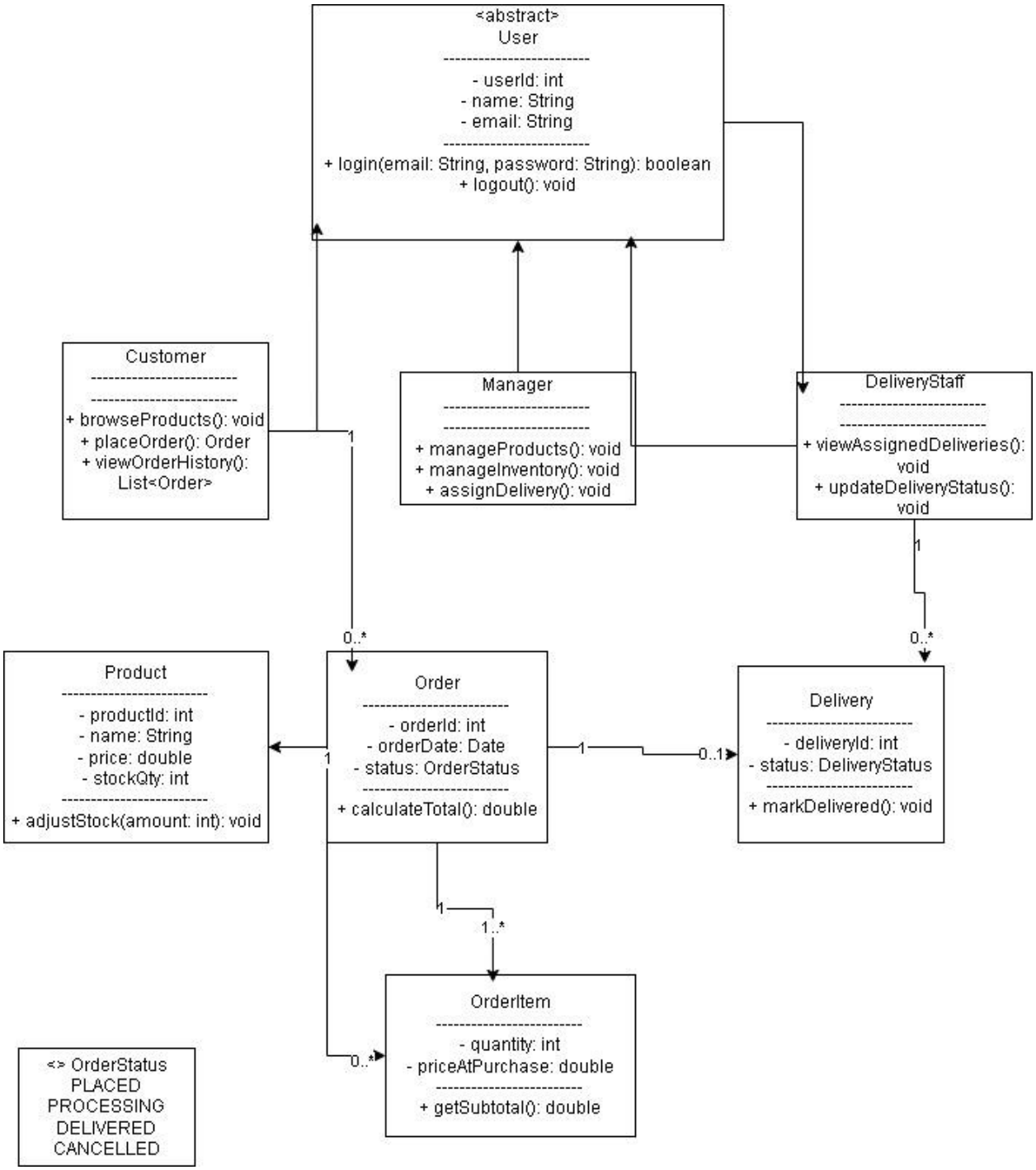


Figure: Grocery Delivery Management System Class Diagram

Grocery Delivery Management System

Module 6 – System Sequence Diagrams and Activity Diagrams

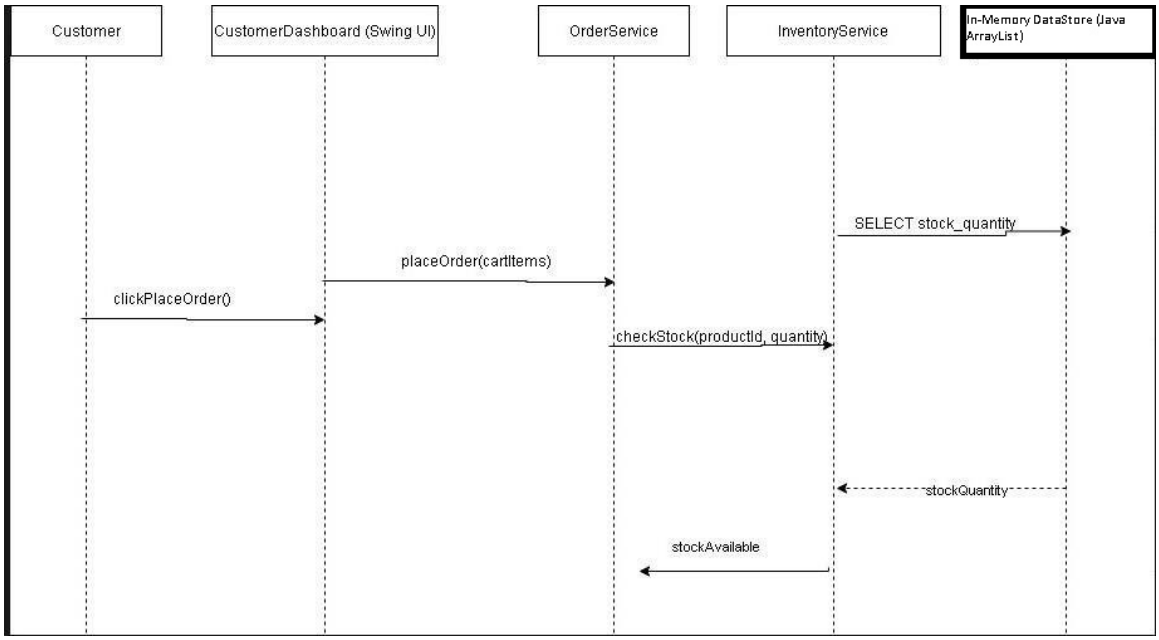
Susan Rogers

Introduction

This document communicates the design solutions and implementation approaches for the Grocery Delivery Management System. It presents UML system sequence diagrams and UML activity diagrams for two main and complex use cases, in accordance with the assignment requirements.

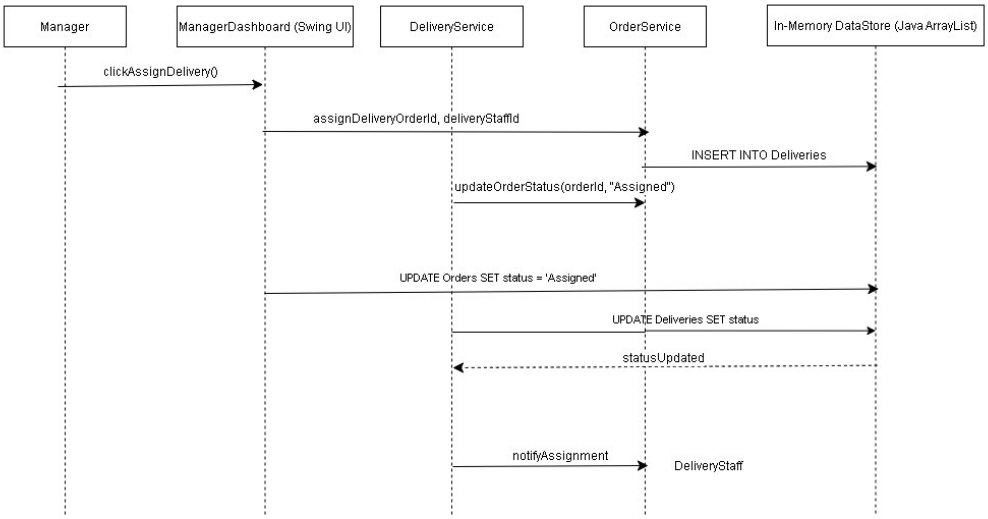
System Sequence Diagram – Use Case 1: Place Grocery Delivery Order

This sequence diagram illustrates how a customer places a grocery delivery order. It shows interactions between the Customer, CustomerDashboard (Swing UI), OrderService, InventoryService, and In-memory DataStore (Java ArrayList).



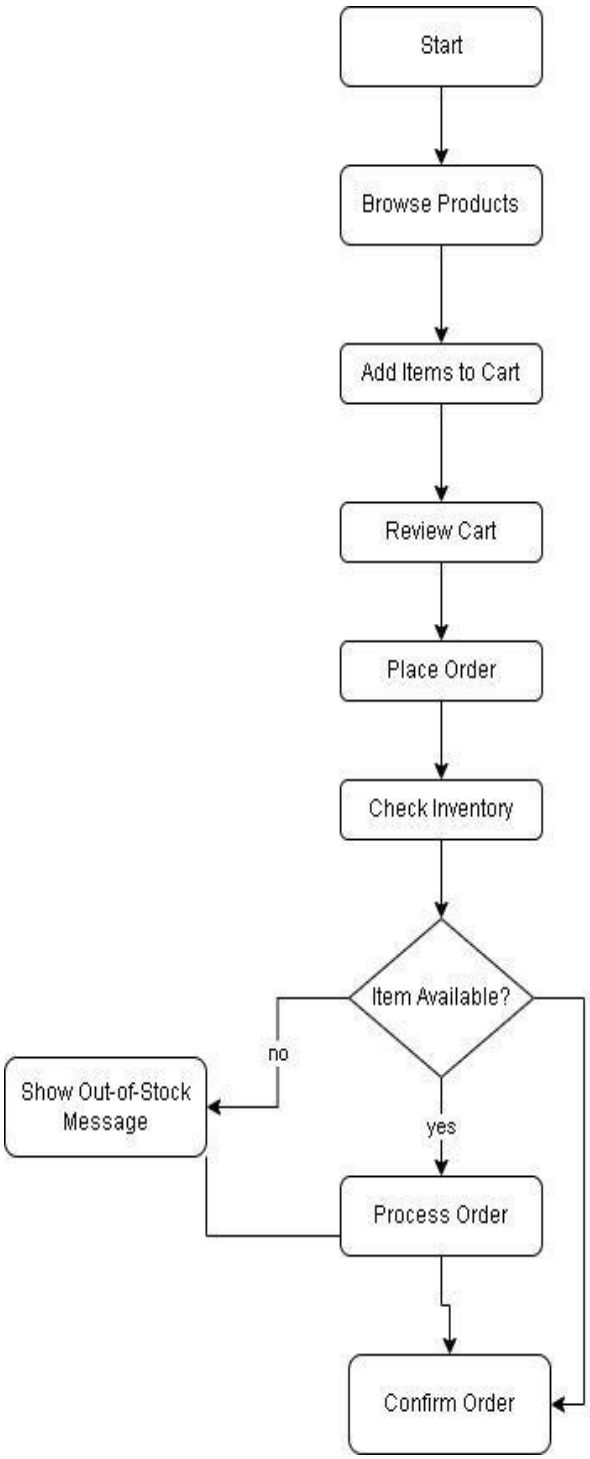
System Sequence Diagram – Use Case 2: Assign Delivery / Track Delivery

This sequence diagram illustrates how a manager assigns delivery staff to an order and how delivery status updates are handled within the system.



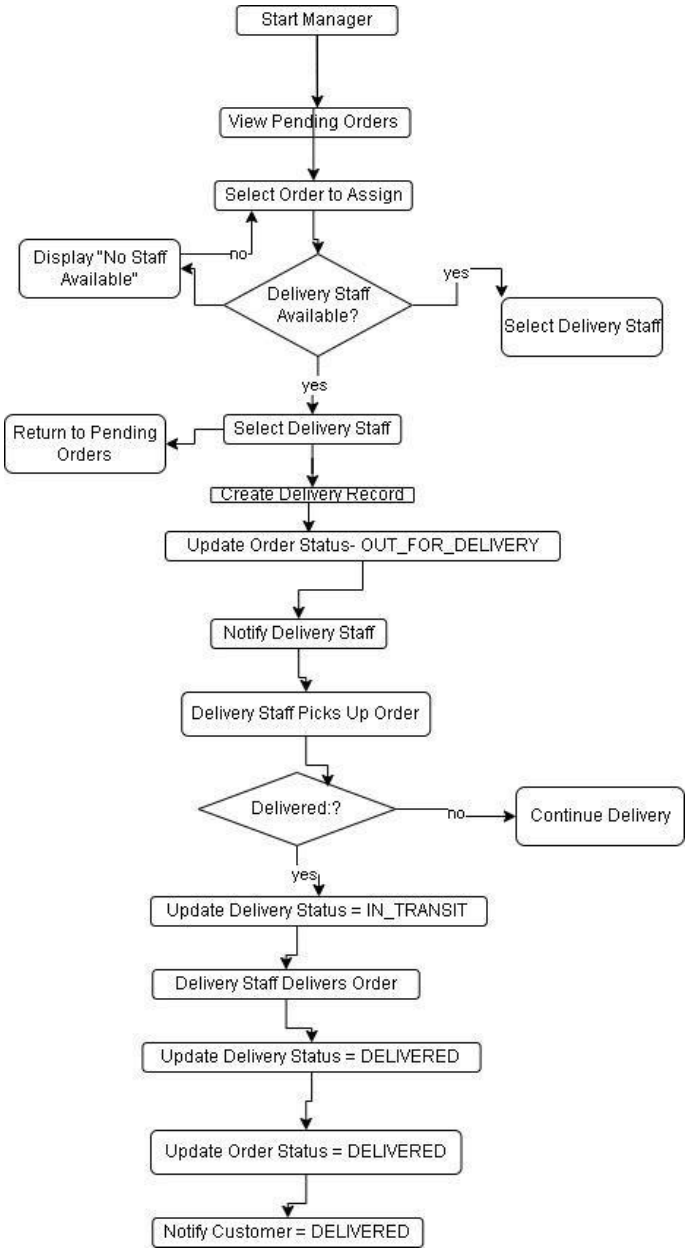
Activity Diagram – Use Case 1: Place Grocery Delivery Order

This UML activity diagram represents the workflow for placing a grocery delivery order, including browsing products, adding items to the cart, inventory validation, order processing, and order confirmation.



Activity Diagram – Use Case 2: Assign Delivery / Track Delivery

This UML activity diagram shows the workflow for assigning delivery staff, tracking delivery progress, updating delivery and order statuses, and notifying relevant parties.



Conclusion

This document provides a clear representation of the system design and workflow for the Grocery Delivery Management System. The included UML sequence and activity diagrams demonstrate how the system supports core grocery ordering and delivery processes.

Grocery Delivery Management System

User Interface Specification

Susan Rogers

Purpose of This Document

This document describes how people use the Grocery Delivery Management System from a user interface perspective. It explains what users see on the screen and how they move through the system while completing everyday tasks. No programming code or technical implementation details are included.
The interface is designed to be straightforward and easy to navigate. Customers, managers, and delivery staff should be able to complete their work with minimal effort and without unnecessary confusion. The interface designs described here are based on the system requirements established earlier in the project.

2. Selected Use Cases for User Interface Design

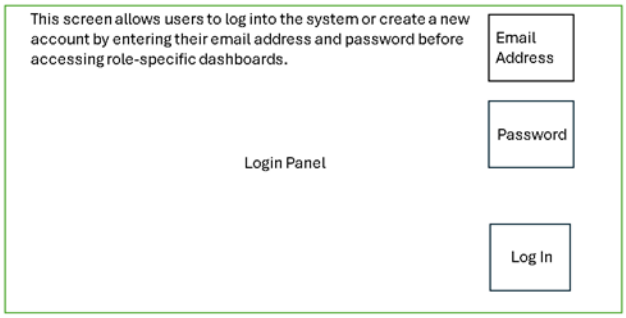
We selected these five use cases because they cover the most common and important ways users interact with the system:

- Register / Log In
- Browse and Search Products
- Place Grocery Delivery Order
- Assign Delivery Order
- Track Delivery Status

Together, these use cases represent the core functions of the Grocery Delivery Management System and reflect the workflows discussed in earlier design materials.

3. Preliminary User Interface Design

3.1 Use Case 1: Register / Log In



Primary Actors: Customer, Manager, Delivery Staff
Interface Description:
The login screen appears when a user first accesses the system. Before using any system features, users are required to enter valid login information. This screen provides the following options:

- Email address input field
- Password input field
- "Log In" button
- "Create Account" link

New users go to a registration screen where they enter the following information:

- Name
- Email address
- Password
- Password confirmation
- User role (Customer, Manager, or Delivery Staff)

Navigation path:
Login Screen → User Dashboard (based on assigned role)
System feedback:

- A message is displayed when a user enters incorrect login information
- A confirmation message is shown once the user successfully signs in

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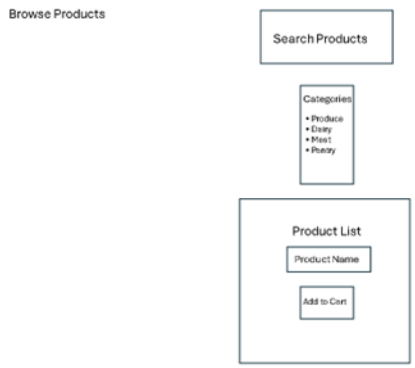
3.2 Use Case 2: Browse and Search Products



Primary Actor: Customer
Interface Description:
The product browsing screen lets customers view grocery items that are available for ordering. Items are organized by category to help users find products more easily. Each product listing includes:

- Product name
- Price
- Available quantity
- An "Add to Cart" button

A search bar at the top lets customers find items by typing keywords.
Navigation path:
Customer Dashboard → Browse Products → Product List
System Feedback: Search results update as the customer enters search terms

- An "Out of Stock" message is shown when an item cannot be ordered

3.3 Use Case 3: Place Grocery Delivery Order

Primary Actor: Customer

Interface description:
The shopping cart screen displays all items the customer has selected. This screen includes:

- A list of selected grocery items
- Quantity controls for each item
- Individual item subtotals
- The overall order total
- A “Place Order” button

After submitting the order, the confirmation screen shows:

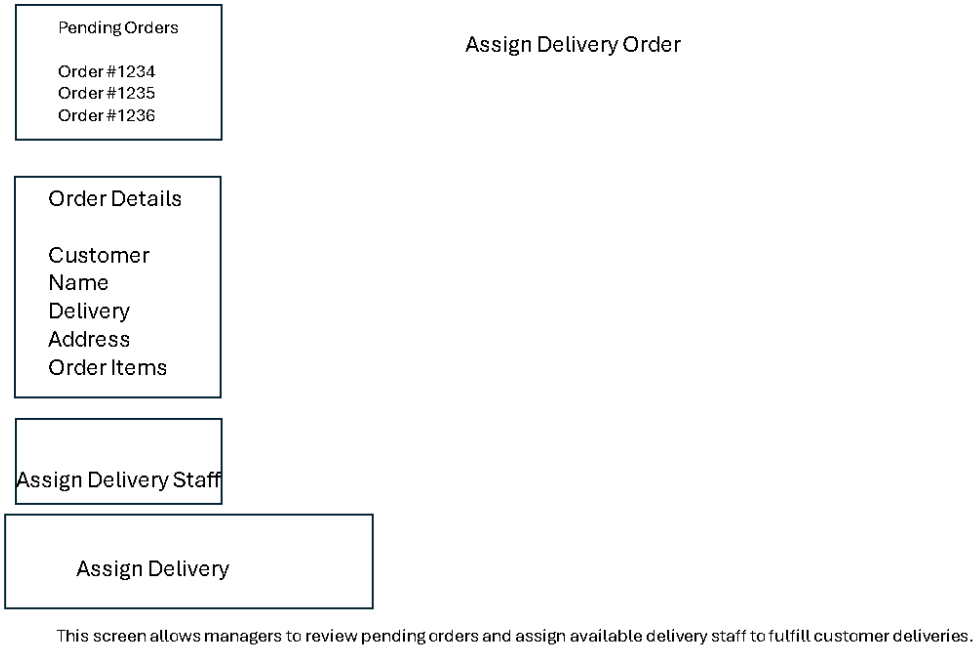
- A unique order number
- Current delivery status
- Estimated delivery time

Navigation Path:
Browse Navigation path: Cart → Place Order → Order Confirmation
System feedback:

- A message confirming that inventory is available
- A notification indicating that the order was successfully placed

This interface follows the order processing flow shown in the system’s activity and sequence diagrams.

3.4 Use Case 4: Assign Delivery Order



Primary Actor: Manager

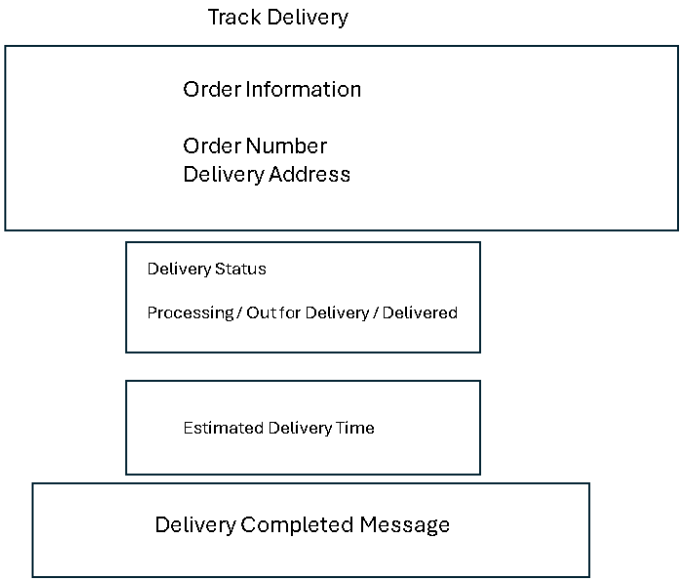
Interface Description: The dashboard displays all orders waiting to be assigned. When an order is selected, a delivery assignment screen appears showing:

- Order details
- A list of available delivery staff
- An “Assign Delivery” button

Navigation path:
Manager Dashboard → Pending Orders → Assign Delivery
System feedback:

- A confirmation message after a delivery assignment is completed
- Automatic update of the order status

3.5 Use Case 5: Track Delivery Status



This screen allows customers to monitor the current status of their grocery delivery and view estimated delivery information.

Primary Actor: Customer
Interface description:
The delivery tracking screen allows customers to check the progress of their orders. The following information is displayed:

- Current delivery status (Processing, Out for Delivery, Delivered)
- Delivery progress updates
- A delivery completion message

Navigation path:
Customer Dashboard → Order History → Track Delivery
System feedback:

- Notifications when the delivery status changes
- A final confirmation when delivery is completed

4. User Effort Estimation

The scenarios below estimate how much effort users need to complete common tasks. Effort is measured by counting mouse clicks and keystrokes for each scenario.

Scenario 1: Customer Places a Grocery Order

Action	Clicks	Keystrokes
Log in	1	15
Search Products	1	8
Add item to cart	1	0
View Cart	1	0
Place order	1	0
Total	5 clicks	23 keystrokes

Scenario 2: Manager Assigns Delivery Order

Action	Clicks	Keystrokes
Log in	1	15
View pending orders	1	0
Select Orders	1	0
Assign delivery staff	1	0
Total	4 Clicks	15 keystrokes

Scenario 3: Customer Tracks Delivery

Action	Clicks	Keystrokes
Log in	1	15
Open order history	1	0
Select Orders	1	0
View status	0	0
Total	3 Clicks	15 keystrokes

5. Conclusion

This User Interface Specification explains how the Grocery Delivery Management System supports user interaction through clear layouts and simple navigation paths. The interface

designs were created to support different user roles while keeping the number of needed actions as low as possible.
The designs here present a practical starting point for future interface development and show how usability was considered throughout the system design.

Grocery Delivery Management System

Traceability Matrix

Purpose

The traceability matrix's goal is to illustrate how each system requirement is addressed within the defined use cases. Its purpose is to ensure that every requirement is accounted for and that every use case is linked to a specific system need. Following this way helps maintain a clear and organized system design which avoids unnecessary or unsupported functionality. Additionally, the matrix highlights which use cases are linked to higher-priority requirements. These priority values assist in planning the order of feature development and in determining which use cases should be prioritized during the final system demonstration.

System Requirements

Req No.	Priority Weight (1–5)	Description
REQ1	5	Users shall be able to register and log in securely
REQ2	5	Customers shall be able to browse grocery products
REQ3	4	Customers shall be able to search for products
REQ4	5	Customers shall be able to add items to a shopping cart
REQ5	5	Customers shall be able to place grocery delivery orders
REQ6	4	Customers shall be able to view order history
REQ7	4	Customers shall be able to track delivery status
REQ8	5	Managers shall be able to add, update, or remove products
REQ9	4	Inventory shall automatically update when orders are placed
REQ10	4	Managers shall be able to assign delivery orders
REQ11	3	Delivery staff shall be able to update delivery status
REQ12	3	Customers shall receive system notifications

Use Cases

UC No.	Description
UC1	Register / Log In
UC2	Browse Products

UC No.	Description
UC3	Search Products
UC4	Add Items to Cart
UC5	Place Grocery Delivery Order
UC6	View Order History
UC7	Track Delivery Status
UC8	Manage Products
UC9	Update Inventory
UC10	Assign Delivery Order
UC11	Update Delivery Status
UC12	Receive Notifications
UC13	Log Out

Traceability Matrix

Req \ UC	UC1	UC2	UC3	UC4	UC5	UC6	UC7	UC8	UC9	UC10	UC11	UC12	UC13
REQ1 (5)	X												X
REQ2 (5)		X											
REQ3 (4)			X										
REQ4 (5)				X									
REQ5 (5)					X				X			X	
REQ6 (4)						X							
REQ7 (4)							X					X	
REQ8 (5)								X					
REQ9 (4)					X				X				
REQ10 (4)										X			
REQ11 (3)											X		
REQ12 (3)					X		X					X	

Priority Weight Summary (Same Logic as Sample)

Use Case	UC1	UC2	UC3	UC4	UC5	UC6	UC7	UC8	UC9	UC10	UC11	UC12	UC13
Max PW	5	5	4	5	5	4	4	5	4	4	3	3	5
Total PW	5	5	4	5	17	4	11	5	8	4	3	6	5

Implementation Priority Discussion

In relation to the traceability matrix, Use Case 5: Place Grocery Delivery Order is the biggest total priority weight. This use case supports several high-priority requirements, including order placement, inventory updates, delivery tracking, and notifications. Because of its central role in the system, this use case should be implemented and demonstrated first.

Other high-priority use cases include Register / Log In, Browse Products, Manage Products, and Track Delivery Status, as they support essential system functionality for customers and managers. Lower-priority use cases, such as updating delivery status and notifications, are still necessary but can be scheduled after the core workflows are complete.

References

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Draw.io Diagram Tool

https://app.diagrams.net